

ADD-ON USB POWER CIRCUIT For UPS

A. SAMIUDHIN

Most computer UPS have only 230V output and no USB output. Presented here is a circuit that gives 5V USB output using the 12V, 7Ah UPS battery fitted inside a UPS. With this circuit you can power USB gadgets directly from a UPS without turning on the PC. This circuit can also be used to power development boards like Arduino and Raspberry Pi.

Circuit and working

As shown in Fig. 1, the circuit is built around switch-mode regulator LM2576-5V (IC1). The LM2576 series offers a high-efficiency replacement for popular three-terminal linear regulators. It substantially reduces the size of the heat-sink; in some cases, no heat-sink is required. It provides all the active functions of a step-down (buck) switching regulator, capable of driving up to 3A load with excellent line and load regulation. The 5V version of LM2576 (LM2576-5V) is used in this circuit.

The 12V battery is connected to CON1. IC1, along with inductor L1 and diode D1, steps down and converts the 12V input into 5V output. Capacitor C2 reduces ripples in the output. Feedback pin 4 of IC1 is directly connected to the output at C2.

IC1 can be turned on/off by an external control signal or else it can be turned on permanently by grounding pin 5 of IC1 through switch S1. When S1 is in position A, IC1 can be controlled by connecting an external control signal at CON3. When S1 is in position B, IC1 is switched on permanently. LED1 indicates whether IC1 is on or off. This circuit can be used to power gadgets digitally.

Construction and testing

An actual-size PCB layout for the add-on USB power circuit for a UPS is shown in Fig. 2 and its components

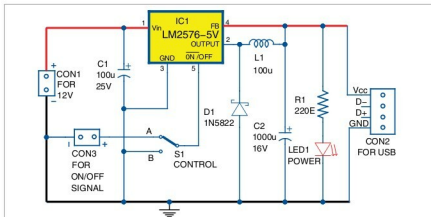


Fig. 1: Circuit diagram of add-on USB power circuit for UPS

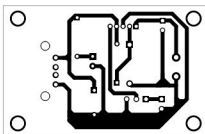


Fig. 2: Actual-size PCB layout of add-on USB power circuit for UPS

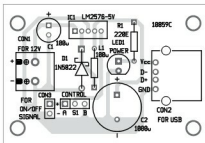


Fig. 3: Components layout for the PCB

layout in Fig. 3. Assemble the circuit on the PCB and place it at a suitable location inside the UPS.

Use suitable heat-sink for IC1. Inductor L1 must have a minimum current rating of 1A. Capacitor C2 should have low equivalent series resistance (ESR) for better perfor-

PARTS LIST

- Semiconductors:**
- IC1 - LM2576-5V voltage regulator
 - D1 - 1N5822 Schottky diode
 - LED1 - 5mm LED
- Resistors (all 1/4-watt, ±5% carbon):**
- R1 - 220-ohm
- Capacitors:**
- C1 - 100µF, 25V electrolytic
 - C2 - 1000µF, 16V electrolytic
- Miscellaneous:**
- CON1 - 2-pin terminal connector
 - CON2 - USB A-type connector
 - CON3 - 2-pin connector
 - L1 - 100µH inductor
 - S1 - SPDT switch
 - 12V, 7Ah UPS battery
 - USB cable
 - Heat-sink for IC1

mance. Any voltage from 7-40V can be used for the circuit.

If the external control signal is open-collector type, a suitable pull-up resistor must be used on pin 5 of IC1. Do not leave this pin float. To maintain stability, capacitor C1 leads must be kept short and placed near IC1. **EFY**



A. Samiuddin takes keen interest in LED lighting, power electronics, microcontrollers and Arduino programming